

3.5.2 Coordination may be chemical or electrical in nature.

Principles	Nerve cells pass electrical impulses along their length. They stimulate their target cells by secreting chemical neurotransmitters directly on to them. This results in rapid, short-lived and localised responses. Mammalian hormones are substances that stimulate their target cells via the blood system. This results in slow, long-lasting and widespread responses. Histamine and prostaglandins are local chemical mediators released by some mammalian cells and affect only cells in their immediate vicinity. In flowering plants, specific growth factors diffuse from growing regions to other tissues. They regulate growth in response to directional stimuli. The role of indoleacetic acid (IAA) in controlling tropisms in flowering plants.
Nerve impulses	The structure of a myelinated motor neurone. The establishment of a resting potential in terms of differential membrane permeability, electrochemical gradients and the movement of sodium and potassium ions. Changes in membrane permeability lead to depolarisation and the generation of an action potential. The all-or-nothing principle. The passage of an action potential along non-myelinated and myelinated axons, resulting in nerve impulses. The nature and importance of the refractory period in producing discrete impulses. Factors affecting the speed of conduction: myelination and saltatory conduction; axon diameter; temperature.
Synaptic transmission	The detailed structure of a synapse and of a neuromuscular junction. Candidates should be able to explain • unidirectionality • temporal and spatial summation • inhibition. The sequence of events involved in transmission across a cholinergic synapse and across a neuromuscular junction. When provided with information, candidates should be able to predict and explain the effects of specific drugs on a synapse. Recall of the names and mode of action of individual drugs will not be required.